Domen nomlari tizimi

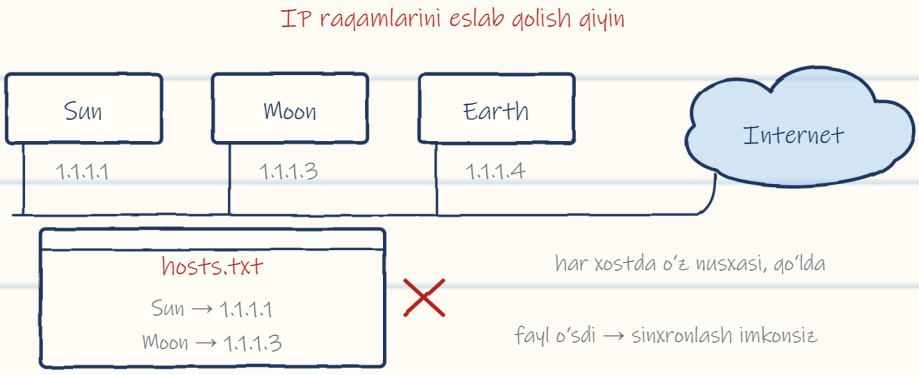
DNS — Domain Name System

Asosiy atamalar

- Domen nomi — xostning odam o'qiydigan ierarxik nomi
- FQDN — ildizgacha to'liq yozilgan nom, nuqta bilan tugaydi
- DNS server — nomni IP manzilga aylantirib beradi
- Zona — bitta server javob beradigan nom fazosi bo'lagi
- Rekursiv so'rov — server javobni o'zi topib beradi

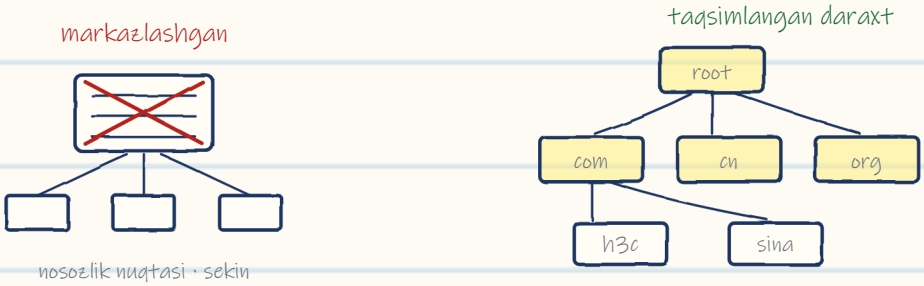
1. Nega nom kerak?

IP manzil raqam — esda qolmaydi. Nom odam uchun, IP mashina uchun.



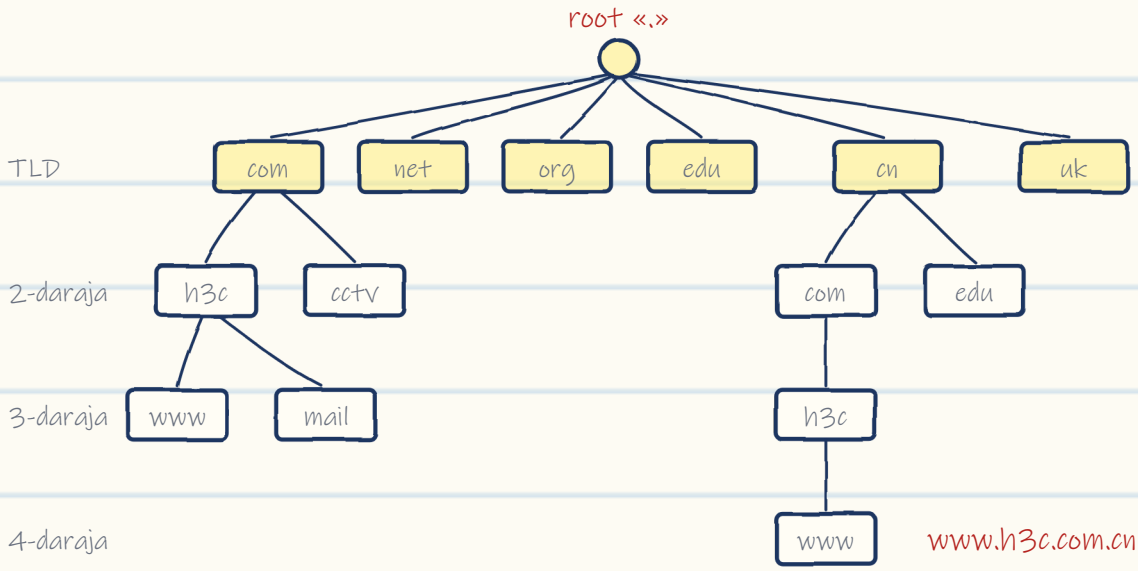
1-rasm. Lokal hosts.txt — 1980-yillarda tugagan yo'l

- vazifa — nom ↔ IP tarjimasi
- ish tartibi — klient-server, UDP/TCP port 53
- tuzilma — daraxtsimon taqsimlangan baza



2. Domen nomi tuzilishi

Chapda — eng aniq, o'ngda — eng umumiy. Nuqta bilan ajratiladi, registr farq qilmaydi.



nuqta bilan tugagan to'liq nom — FQDN

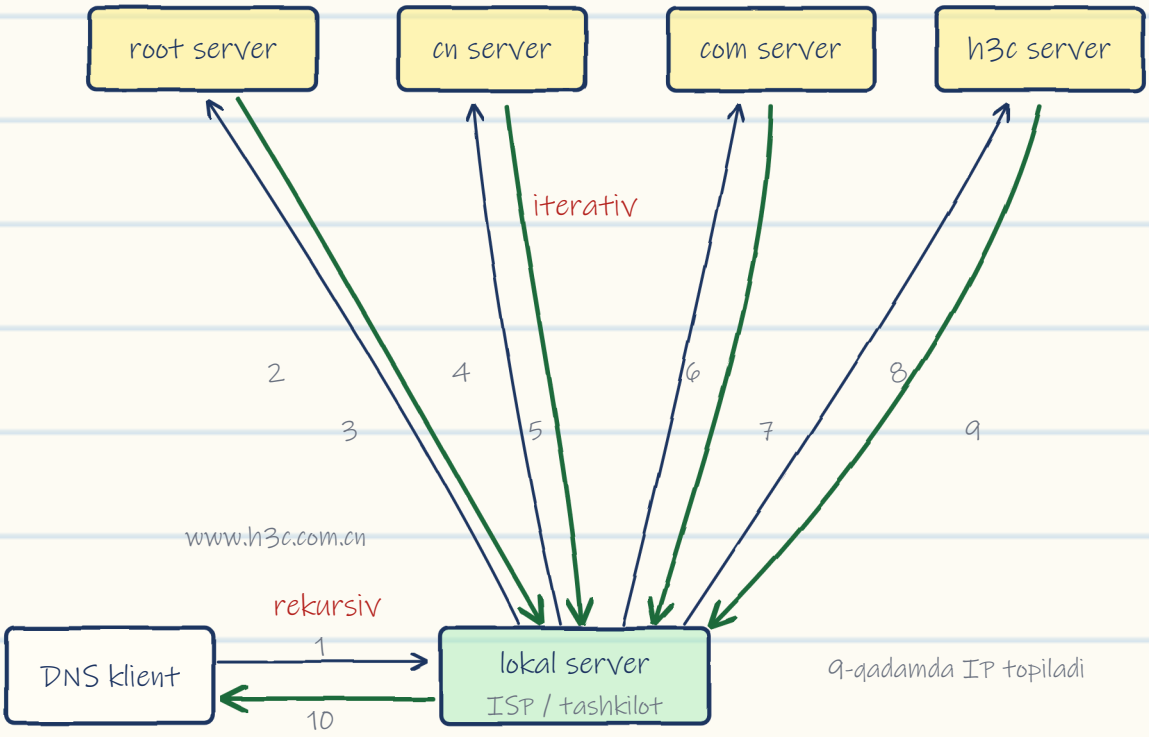
2-rasm. Teskari daraxt: root → TLD → 2-daraja → ...

TLD turi	misol
davlat (ISO 3166)	cn · us · uk — ~200 ta
xalqaro	int
umumiy	com net org edu gov mil + 2000-yil aero biz coop info museum name pro

Uzunlik: har daraja ≤ 63 bayt, to'liq nom ≤ 255 bayt

3. Nom yechilishi

Dastur resolverni chaqiradi; yechim yuqoridan pastga boradi, lekin ishni asosan lokal server bajaradi.



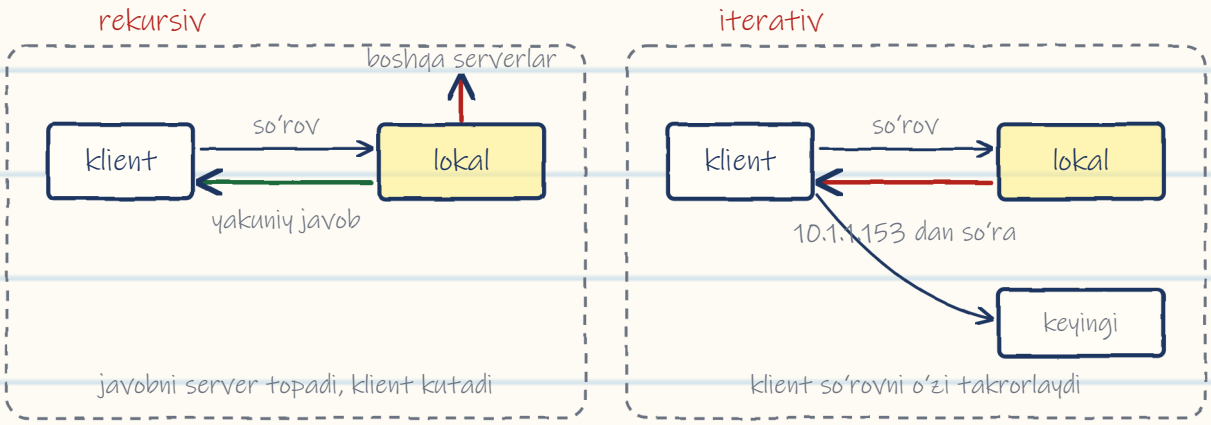
3-rasm. 10 qadam: klient → lokal → root → cn → com → h3c

- lokal server — klientga yaqin, ko‘p so‘rovni o‘zi yopadi
- root server — nom yechmaydi, faqat TLD serverlari IP’sini beradi
- resolver — klientda ham, serverda ham nom yechuvchi modul

Qoida: yechim yuqoridan pastga, ammo so‘rovlarning ko‘pini lokal server o‘zi yopadi

4. So‘rov rejimlari

Klient → lokal server: odatda rekursiv. Lokal server → root: odatda iterativ.



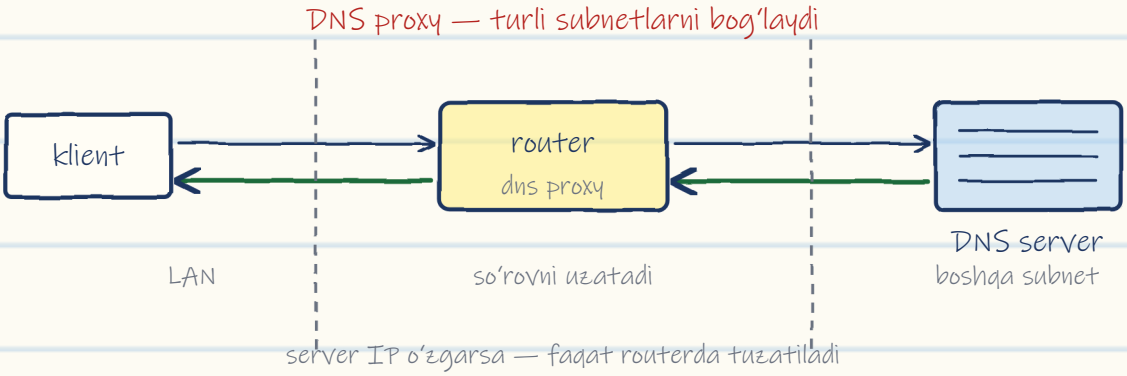
4-rasm. Rekursiv — «javobni topib beraman», iterativ — «boshqadan so'ra»

5. Teskari so'rov



5-rasm. in-addr.arpa — IP maxsus domen nomi sifatida

6. H3C qurilmasida DNS



statik	ip host *hostname ip-address*
dinamik	dns server *ip-address*
suffiks	dns domain *domain-name*
proksi	dns proxy enable
ko'rish	display ip host · display dns server · display dns domain

Xulosa

- DNS — nom ↔ IP, klient-server, port 53, UDP/TCP
- Nomlar fazosi — teskari daraxt: root → TLD → 2-daraja → …
- Lokal server ko'p ishni bajaradi, root faqat yo'l ko'rsatadi
- Rekursiv — server topib beradi; iterativ — klient takrorlaydi
- Teskari so'rov — in-addr.arpa, PTR yozuvi

Tekshirish

1. DNS porti? — 53, UDP va TCP
2. To'liq nom uzunligi? — ≤ 255 bayt, daraja ≤ 63
3. Root server nima qiladi? — TLD serverlari IP'sini beradi
4. 192.168.1.20 teskari nomi? — 20.1.168.192.in-addr.arpa